

Sampling I

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Outline

- Key sampling vocabulary
- Probability Sampling



Key Sampling Vocabulary



Target Population

- The target population is the group of units to which survey results will be generalized.
- It includes not just individuals, but entities like households, schools, hospitals, businesses, and more.
- The goal is to ensure that the survey results apply to this broader population.



Element

- Each member of the population
- Note:
 - Not all elements are sampled!



List Sample

- List of the individual elements in the entire population
- Not always available 😊



Sampling Frame

- A list, map, or other specification of elements in the population from which a sample may be selected.
- Examples include a list of all university students, a telephone directory, or a map of geographic areas.



Sampling unit

- For the purpose of sample selection, the population should be divisible into a finite number of distinct, nonoverlapping and identifiable units called **sampling units**, so that each member of the population belongs to only one sampling unit.
- A sampling unit refers to an element or a set of elements about which information is measured.



Observation/Case/Subject

- Each selected element for which data were gathered successfully



Coverage

- Proportion of the target population that can be selected & surveyed
- Often expressed as a percentage (%)



Defects in Coverage

- Called non-coverage/under coverage
- Arise because we usually can't survey every element of our target population
 - Practical limitations (time, cost)
 - Information about the population (we can't survey people we don't know about or can't reach)



Problems arising from Undercoverage

- Undercoverage is usually not random
 - Often **systematically related** to participant characteristics
- Consequence: Sample may not accurately represent all subgroups within the population



Probability Samples

- Each element in the population has a known probability of being selected
- Types of probability samples
 - Simple random sample
 - Stratified sample
 - Cluster sample
 - Systematic sample



- Most results in sampling rely on the sampling distribution



An example

- We are interested in the average number of hours UofM undergraduates spend studying during the study week
- What are the population and parameter?
- A random sample of 10 UofM undergraduates:
 - 10, 5, 13, 10, 20, 22, 12, 8, 4, 16
- What are the sample and statistic?

An example

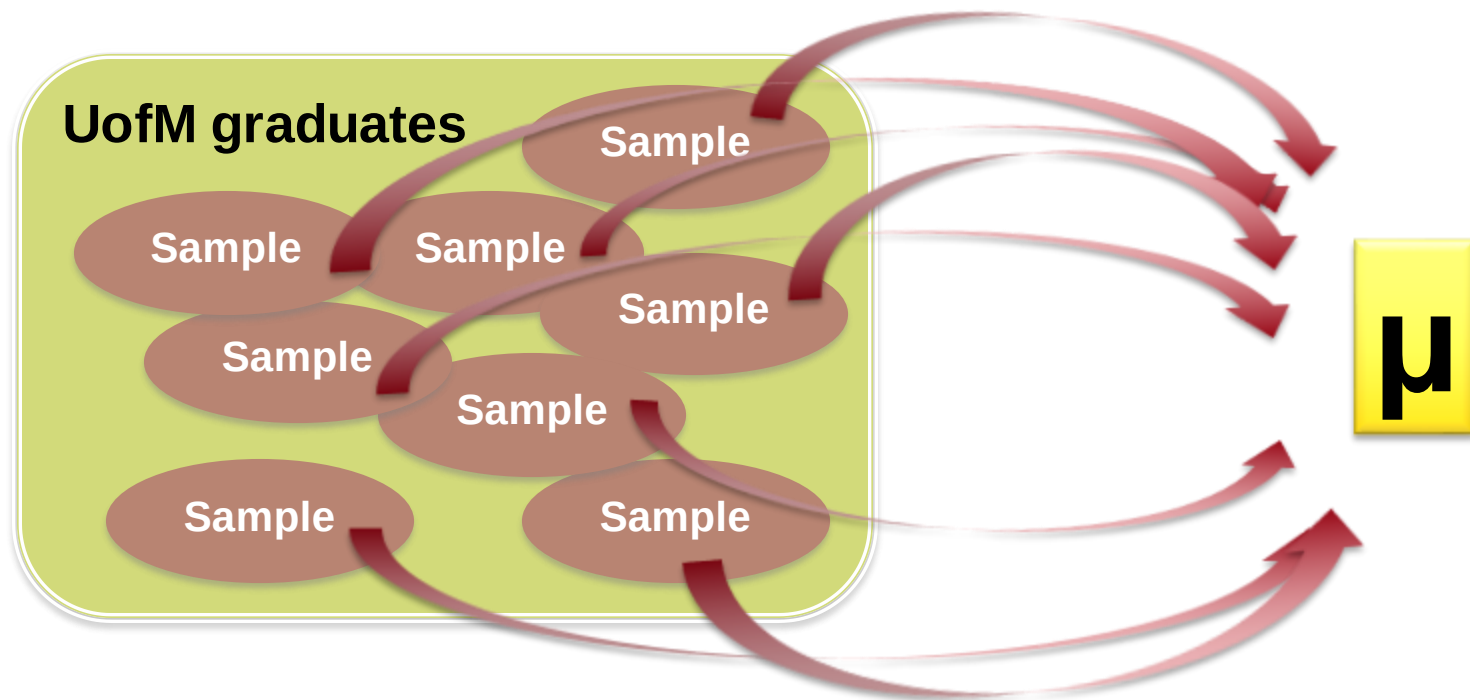
- Population: all UofM undergraduates
- Parameter (μ): the average number of hours spend studying during finals week
- Sample: 10 UofM undergraduates
- Statistic ($\bar{M}$): 12
- Conclusion: The estimated average number of hours all UofM undergraduates spend studying during finals week is 12.

An example

- How accurate is the estimate?
- How do we assess the accuracy of our estimate?

Sampling Variability

What would happen if we took many samples of same size?



Sampling Distributions

If we took every one of the possible samples of a certain size, calculated the sample mean for each, and graphed all of those values, we'd have a **sampling distribution of the sample mean**.

The **population distribution** of a variable is the distribution of values of the variable among all individuals in the population.

The **sampling distribution** of a statistic is the distribution of values taken by the statistic in all possible samples of the same size from the same population.



UofM graduates

Sample

Sample

Sample

Sample

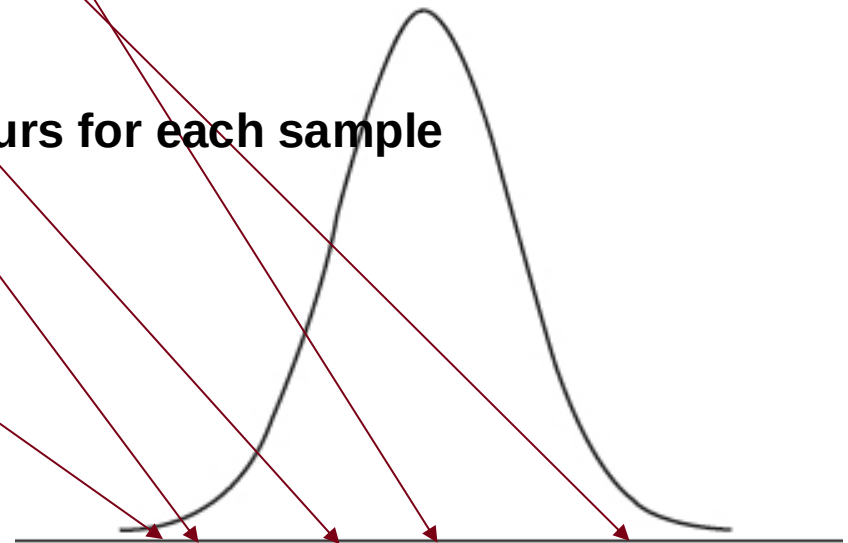
Sample

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Sample

Calculate average number of hours for each sample

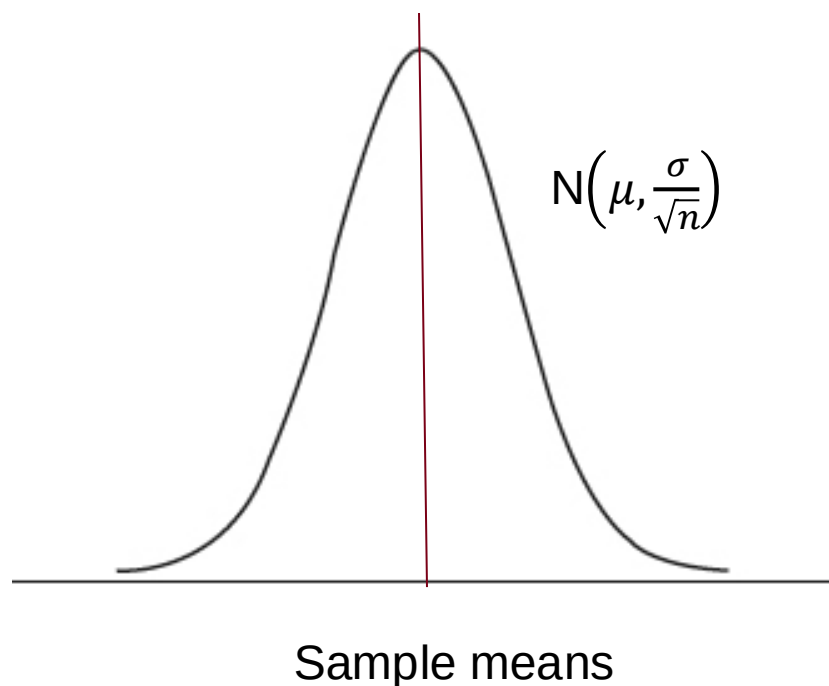


Sample means

- The standard deviation of a sampling distribution is called the **standard error**

$$SE = \frac{\sigma}{\sqrt{n}}$$

- Standard Error
 - The extent to which sample means selected from the same population differ from one another
 - How accurate the sample mean is



Factors That Decrease Standard Error

We want to have an accurate sample mean (smaller standard error).

What can we do to achieve this goal?

$$s_M = \sqrt{\frac{s^2}{n}}$$

Because we sometimes use biased estimators, we often use the mean squared error (MSE) rather than variance to measure the accuracy of an estimator:

$$\begin{aligned}\text{MSE}[\hat{t}] &= E\left[(\hat{t} - t)^2\right] \\ &= E\left[(\hat{t} - E[\hat{t}] + E[\hat{t}] - t)^2\right] \\ &= E\left[(\hat{t} - E[\hat{t}])^2\right] + (E[\hat{t}] - t)^2 + 2E\left[(\hat{t} - E[\hat{t}])(E[\hat{t}] - t)\right] \\ &= V(\hat{t}) + [\text{Bias}(\hat{t})]^2.\end{aligned}$$

An estimate is unbiased if $E(\hat{t}) = t$, precise if $V(\hat{t})$ is small, and accurate if $\text{MSE}(\hat{t})$ is small.



Simple Random Samples

- Uses random selection
- Each element has the same probability of being included



Simple Random Samples

- To take an SRS, you need a list of all observation units in the population; this list is the sampling frame.
- In an SRS, the sampling unit and observation unit coincide.



Simple Random Samples

- Whether one element is selected has no effect on whether any other element is selected
 - Selections are *independent*
- Usually conducted *without replacement*
 - After an element is selected it can't be selected again



Simple Random Samples

- Estimating a population mean

$$\hat{y} = \bar{y}$$

- Estimating population variance

$$s^2 = \frac{1}{n-1} \sum (y_i - \bar{y})^2$$



Simple Random Samples

- Variance of the sample mean is

$$V(\bar{y}) = \frac{S^2}{n} \left(1 - \frac{n}{N}\right)$$

- Factor $(1 - n/N)$ is called the finite population correction (fpc).



Sampling weight

- Sampling weight of sampled unit i can be interpreted as the number of population units represented by unit i .
- Sometimes called the design weight
- The inclusion probability in SRS is

$$\pi_i = \frac{n}{N}$$

- The sampling weight sampled unit i is the reciprocal of inclusion probability

$$w_i = \frac{1}{\pi_i}$$



Sampling weight

- The population size is estimated by the sum of the weights for the sample:

$$N = \sum_i w_i$$

- the population mean is estimated by

$$\hat{y} = \bar{y} = \frac{\sum_i w_i y_i}{\sum_i w_i}$$



Sampling distribution under SRS

- Many times, we are not only interested in estimating population mean
- We also report the accuracy of the estimation
- The standard error (SE) is the square root of the estimated variance of sample mean

$$\text{SE}(\bar{y}) = \sqrt{\left(1 - \frac{n}{N}\right) \frac{s^2}{n}}$$



Sampling distribution under SRS

- In addition to point estimation, we might be also interested in interval estimation
- We can report confidence interval



UofM graduates

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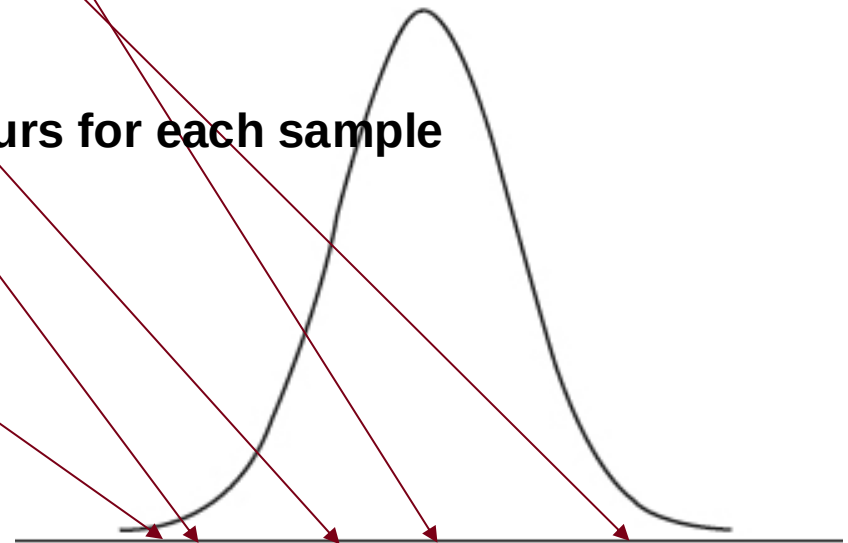
Sample

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Sample

Calculate average number of hours for each sample

Q: What are the mean and variance of the sampling distribution?



Sample means

The Sampling Distribution of a Sample Mean

The Sampling Distribution of Sample Means

Suppose that $\bar{M}$ is the mean of a sample of size n drawn from a large population with mean μ and standard deviation σ . Then:

The mean of the sampling distribution of sample mean is *equal to the population mean*

The standard deviation of the sampling distribution of sample mean is

$$SE(\bar{y}) = \sqrt{\left(1 - \frac{n}{N}\right) \frac{s^2}{n}}$$

UofM graduates

Sample

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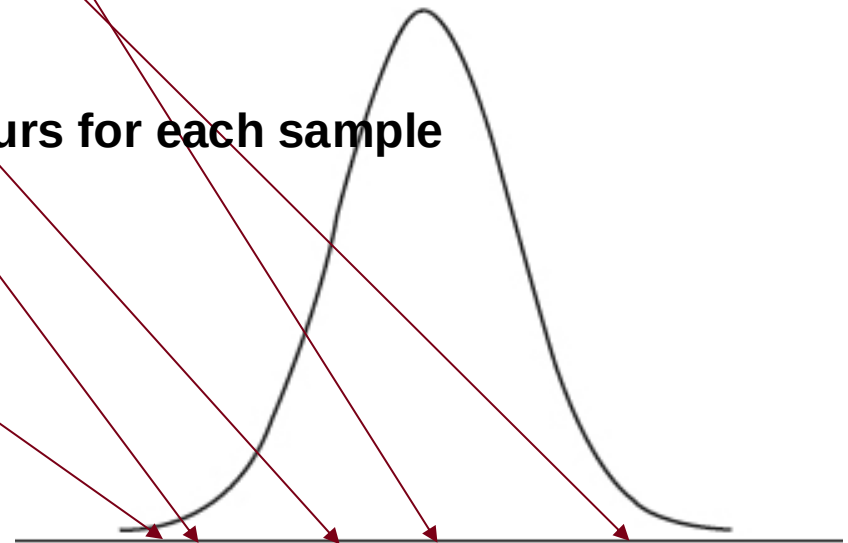
Sample

Sample

Sample

Calculate average number of hours for each sample

Q: What are the shape of the sampling distribution?



Sample means

As the sample size increases, the distribution of sample means begins to look more and more like a Normal distribution!

When the sample is large enough, the distribution of sample means is very close to Normal, *no matter what shape the population distribution has*, as long as the population has a finite standard deviation.

Draw a simple random sample of size n from any population with mean μ and finite standard deviation. The central limit theorem (CLT) says that when n is large, the sampling distribution of the sample mean $\bar{y}$ is approximately Normal:

$$N\left(\mu, \sqrt{\left(1 - \frac{n}{N}\right) \frac{s^2}{n}}\right)$$



An illustration

http://onlinestatbook.com/stat_sim/sampling_dist/

Confidence Intervals

- large-sample $100(1 - \alpha) \%$ CI for the population mean can be calculated as

$$\left[\bar{y} - z_{\frac{\alpha}{2}} \sqrt{\left(1 - \frac{n}{N}\right) \frac{S^2}{n}}, \bar{y} + z_{\alpha/2} \sqrt{\left(1 - \frac{n}{N}\right) \frac{S^2}{n}} \right]$$

where $z_{\frac{\alpha}{2}}$ is the $(1 - \alpha/2)$ th percentile of the standard normal distribution.

$z_{\frac{\alpha}{2}} \sqrt{\left(1 - \frac{n}{N}\right) \frac{S^2}{n}}$ is called margin of error.



Example

- In a Pew Research Center political survey conducted in 2023, 1,500 U.S. adults were polled about their approval of the president's job performance. The survey reported that 52% of respondents approved of the president's performance, with a **margin of error of 3 percentage points** at the **95% confidence level**.
- This means the true approval rating for the president in the general population could range from _____ to _____.



Determining sample size

- Specify the tolerable error

$$P(|\bar{y} - \bar{y}_{\mathcal{U}}| \leq e) = 1 - \alpha$$

- e is the **margin of error**.
- Again, the margin of error is one-half of the width of a CI.

$$e = z_{\alpha/2} \sqrt{\left(1 - \frac{n}{N}\right) \frac{S^2}{n}}$$



Determining sample size

- When N is very large, we can ignore the fpc term and thus

$$n = \left(\frac{S \times z_{\alpha/2}}{e} \right)^2$$

- If we cannot ignore fpc, the sample size is

$$n_{fpc} = \frac{n}{1 + \frac{n-1}{N}}$$



Determining sample size

- For many surveys of people in which a proportion is measured, the margin of error is set at $e = 0.03$ and $\alpha = 0.05$ (Lohr, 2022)
- When interested in a proportion, we can use $1/4$ as an upper bound for S^2 .
- For other quantities, S^2 must be estimated or guessed at.



Simple Random Samples

- Major benefit of simple random samples:
 - Elements are selected with the same probability
 - No weighting required to represent the target population



Exercise I

- Suppose we want to conduct a survey to estimate the proportion of current U of M students who support the use of AI, like ChatGPT, for classroom learning. We plan to take an SRS and want to use a 95% CI with margin of error 0.03.
- Let's first ignore the fpc, the sample size required is _____.
- If we assume U of M has 50,000 students, the sample size required is _____.



Exercise II

- A data set with 2044 observations on 5 variables.
 - Hrsrelax: After an average work day, about how many hours do you have to relax or pursue activities that you enjoy
 - Mntlh1th: For how many days during the past 30 days was your mental health, which includes stress, depression, and problems with emotions, not good?
 - Hrs1: Hours worked each week.
 - Degree: Educational attainment or degree.
 - Grass: Do you think the use of marijuana should be made legal, or not?



Exercise II

- Open gss2010.xlsx file on Canvas using Excel
- Assume the 2044 cases are all units in the sampling frame
- Estimate the proportion of individuals who think the use of marijuana should be made legal with an SRS of size 10, 50 and 100



Exercise II

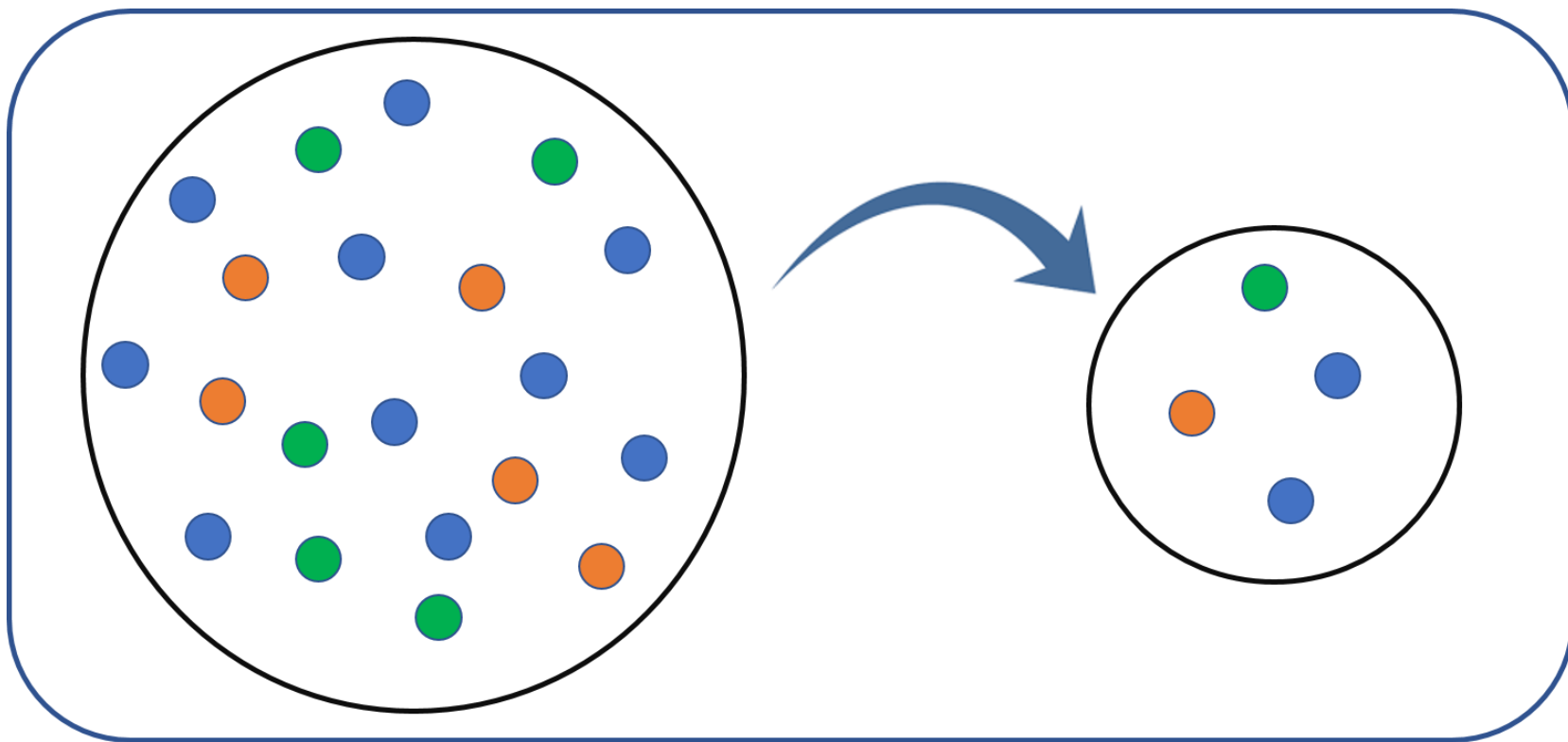
- Now calculate the sample size required to achieve a margin of error 0.03 for 95% CI and draw a sample accordingly
- What is the sampling weight of each individual in the sample?
- What is the inclusion probability of each individual in the sampling frame?
- Calculate the proportion based on the sample.
- Construct a 95% CI

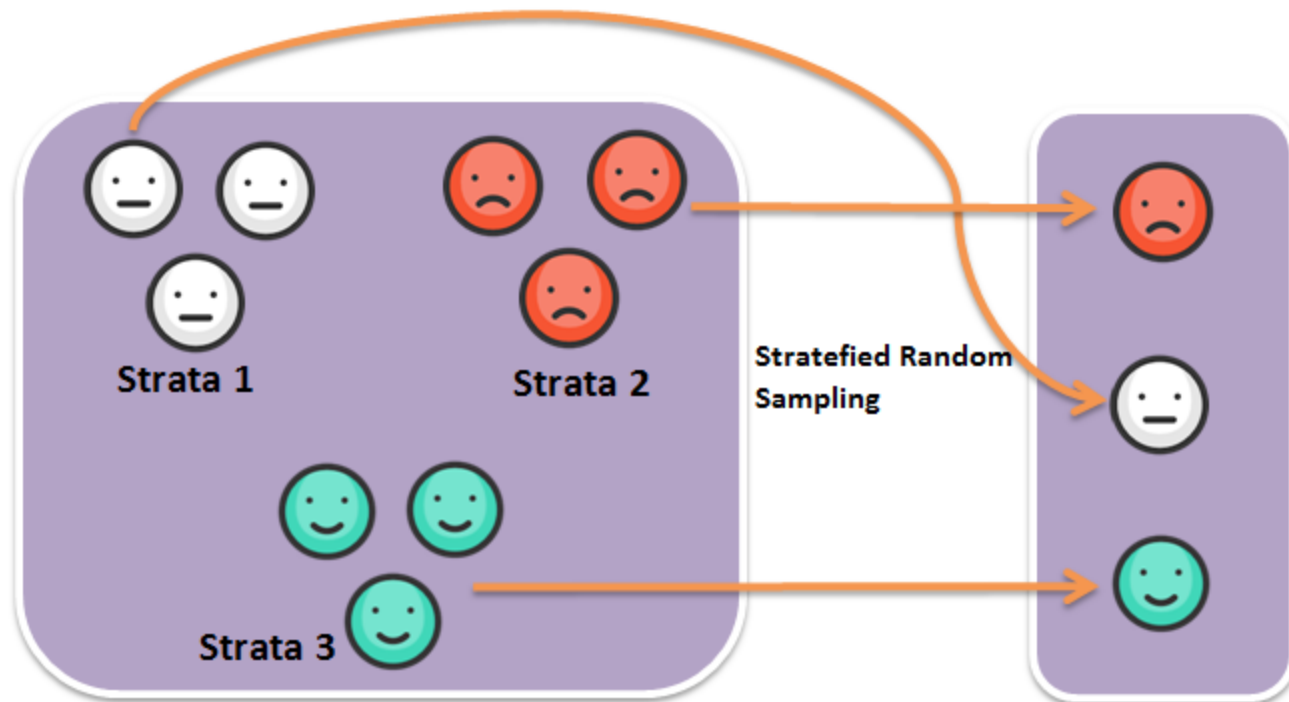


Stratified Random Samples

- Target population is divided into ***strata*** (groups)
 - Groups that don't overlap, but together they represent the target population
 - Simple Random Samples are drawn from each strata separately







An example

The University Health Center is considering the adoption of a new program to help young adults lose weight. Before changing programs, the administration commissioned a survey to find out, among other things, how their new program compared with the current one and how male and female students of different ages performed.

Previous experience had suggested that older students appeared to do better in weight reduction programs. Therefore, the surveyors planned to get a sample of men and women in two age groups—17 to 22 years and 23 to 28 years—and compare their performance in each of the programs.



An example

- About 310 undergraduates signed up for the health center's regular weight reduction program for the winter seminar.
- Of the 310 participants, 140 were between 17 and 22 years old, and 62 of these were men. Also 170 students were between 23 and 28 years, and 80 of these were men.
- The surveyors randomly selected 40 persons from each of the four subgroups (male, female, ages 17–22, and ages 23–28) and randomly assigned every other student to the new program. The sample looked like this:



An example

University Health Center's Weight-Loss Program

	Ages 17–22 Years		Ages 23–28 Years		Total
	Male	Female	Male	Female	
Regular program	20	20	20	20	80
New program	20	20	20	20	80
Totals	40	40	40	40	160

Fink, A. (2015). *How to conduct surveys: A step-by-step guide*. SAGE publications.



Why Stratified Random Samples?

- We want to be protected from the possibility of obtaining a really bad sample.
- We may want data of known precision for subgroups of the population.
- Stratified sampling can be a way to guarantee an exact proportionate representation of the population
- Stratified sampling often gives more precise (having lower variance) estimates for population means and totals.



Practical Example

- Study of high school principals' leadership styles
- Target population includes 80% public schools and 20% private schools
 - Simple Random Sample may not accurately reflect these proportions
- Stratified sample:
 - Select SRSs from public schools & private schools
 - Need to decide if proportional allocation is important



Sampling weight

- Sampling weight of sampled unit i of stratum h can be interpreted as the number of units in stratum h represented by unit i

$$w_{ih} = \frac{1}{\pi_{ih}} = \frac{N_h}{n_h}$$

where π_{ih} is the inclusion probability for unit i .



Sampling weight

- If the population has 1600 men and 400 women, and the stratified sample design specifies sampling 200 men and 200 women, then
- each man in the sample has a weight of _____ and each woman has a weight of _____.



Sampling weight

- The population size is estimated by the sum of the weights for the sample:

$$N = \sum_h \sum_i w_{ih}$$

- the population mean is estimated by

$$\hat{y} = \bar{y} = \frac{\sum_h \sum_i w_{ih} y_{ih}}{\sum_h \sum_i w_{ih}}$$



Proportional Allocation

- Proportional allocation means the same proportion of units was sampled in each stratum
- The number of sampled units in each stratum is proportional to the size of the stratum
- The inclusion probability for unit i in stratum h is the same for all strata.

$$w_{ih} = \frac{N}{n} = \frac{N_h}{n_h}$$

$$\pi_i = \pi_{ih} = \frac{n}{N}$$



Proportional Allocation

- The probability that an individual will be selected to be in the sample, n/N , is the same as in an SRS
- With proportional allocation, each unit in the sample represents the same number of units in the population.



Proportional Allocation

- If the population has 1600 men and 400 women, and the stratified sample design specifies sampling 400 individuals with proportional allocation, then
- The sample will have _____ men and _____ women
- The probability of being selected for each individual is _____
- The sampling weight for men is _____ and for women is _____



Disproportional allocation

- The U.S. Postal Service has conducted surveys asking postal customers about their perceptions of the quality of mail service.
- The population of residential postal service customers is stratified by geographic area, and it is desired that the precision be ± 3 percentage points, at a 95% confidence level, within each area.
- The USPS may choose to sample 1,100 customers in each area.
- This is disproportional but provide desired precision.



Which to use?

- If you are taking a stratified sample in order to ensure that the sample reflects the population with respect to the stratification variable and you would like your sample to be a miniature version of the population, you should use **proportional allocation**.
- **Disproportional allocations** specify oversampling one or more of the strata. It may be needed if you want to guarantee that estimates calculated from individual strata will achieve a specified degree of precision.



Which to use?

- A stratified sample with disproportional allocation is **representative**, in the sense that it can be used to estimate any numerical quantity that could be calculated from the population, but it is **not** a small-scale replica of the population.
- You must use the sampling weights when graphing data and computing estimates; otherwise, your estimates will be biased.



Determining the Total Sample Size

- For proportional allocation, the sample size needed is

$$n_{SRS} \times \frac{\sum_h \frac{N_h}{N} S_h^2}{S^2}$$

- n_{SRS} is the sample size calculated for an SRS.
- $\sum_h \frac{N_h}{N} S_h^2$ is a weighted average of the within-stratum variances and thus will usually be smaller than the overall population variance S^2
- The stratified sampling allows you to achieve a desired precision with a smaller sample size



Determining the Total Sample Size

- If S_h^2 is assumed to be equal to the overall population variance S^2 , for proportional allocation, the sample size for stratum h needed is

$$n_{SRS} \times \frac{N_h}{N}$$

- n_{SRS} is the sample size calculated for an SRS.
- In this case, the stratified sampling has the same sample size as SRS



An example

- Men make up a small minority of early childhood education (ECE) teachers, with an estimate of 3%.
- Now assume that you want to do a survey for ECE teachers in Minnesota to see the proportion of them who are certified.
- Suppose there are about 1,000 ECE teachers in Minnesota and you want to use a 95% CI with margin of error 0.06.
- If you take an SRS, the same size needs to be _____.
- If we do a proportional stratified sample for male and female, the number of male teachers sampled will be _____.



Defining Strata

- Stratified sampling with proportional allocation improves (or does not harm) the precision for every variable y measured in the survey.
- Stratification is most efficient when the stratum means differ widely
- When constructing strata, we want the stratum means for y to be as different as possible.



Defining Strata

- If we want to do a survey to know the salary of early education teachers
- Ideally, H strata with high salary group, ..., low salary group
- What variables may be associated with salary?
 - Experience
 - Race
 - Sex



Oversampling & Undersampling

- Tools for dealing with uneven proportions of relevant subgroups in a target population
 - In our example, there were more public school principals than private school principals



Summary

- Stratified sampling has three major design issues: defining the strata, choosing the total sample size, and allocating the observations to the defined strata.
- proportional allocation yields the same sampling fraction in each stratum.
- Proportional allocation almost always results in smaller variances for estimated means and totals than simple random sampling
- Disproportional allocation may be preferred if some strata should have higher sampling fractions than others, for example, if it is desired to have larger sample sizes for strata with minority populations.



Summary

- Stratified sampling can guarantee a better proportionate representation of the population than SRS
- Stratified sampling often gives more precise (having lower variance) estimates for population means and totals than SRS with the same sample size
- To achieve the same precision, stratified sampling requires smaller sample size than SRS (when each stratum has similar observations, a.k.a smaller within group variances)



Cluster Samples

- First, what is a *cluster*?
 - Naturally occurring group of population elements
 - Examples:
 - Households made up of individuals
 - Schools made up of students
 - Workers within companies



Cluster Samples

- Suppose we want to find out how many bicycles are owned by residents in a community of 10,000 households.
- We could take a simple random sample (SRS) of 400 households
- we could divide the community into 500 blocks of about 20 households each and sample every household in randomly selected 20 blocks --- cluster sampling



Some terms

- The blocks are the **primary sampling units** (psus), or clusters.
- The households are the **secondary sampling units** (ssus)
- Often, the ssus are the elements in the population.

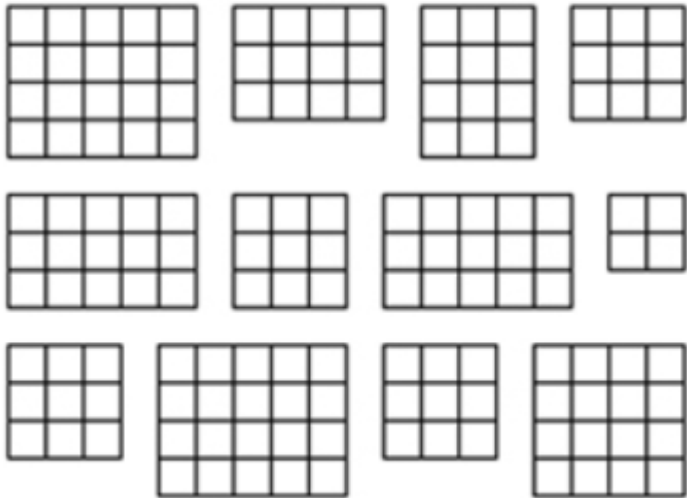
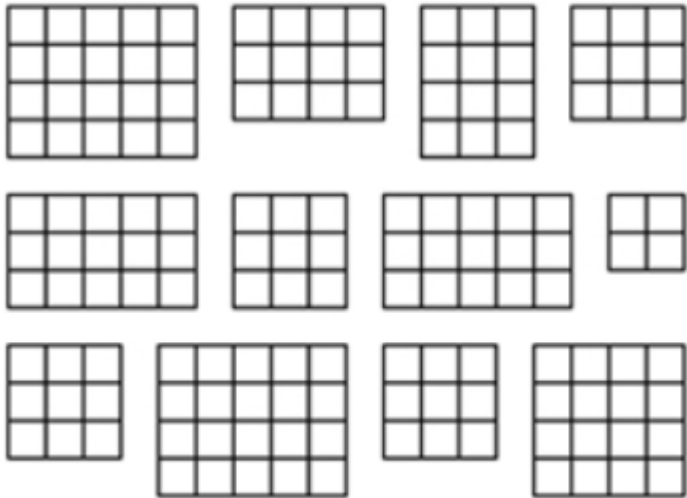


Why use cluster samples?

- Constructing a sampling frame list of observation units may be difficult, expensive, or impossible.
- The population may be widely distributed geographically or may occur in natural clusters such as households or schools, and it is less expensive to take a sample of clusters rather than an SRS of individuals.



Cluster sampling vs stratified sampling

Stratified Sampling	Cluster Sampling
Each element of the population is in exactly one stratum.	Each element of the population is in exactly one cluster.
Population of H strata	Population of N clusters:
	

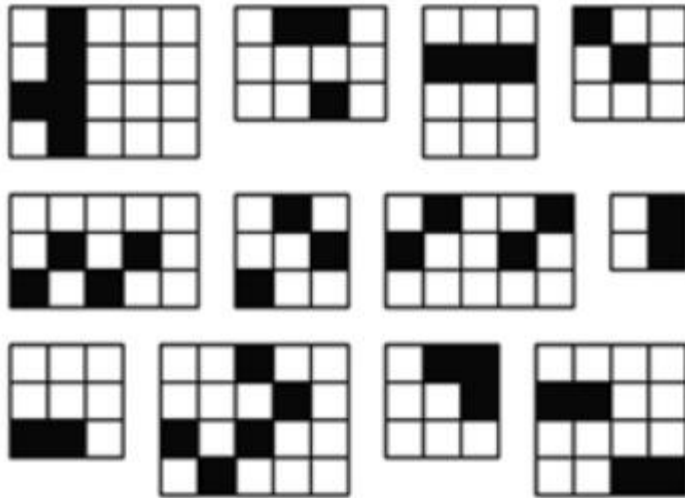


Cluster sampling vs stratified sampling

Stratified Sampling

Take an SRS from *every* stratum.

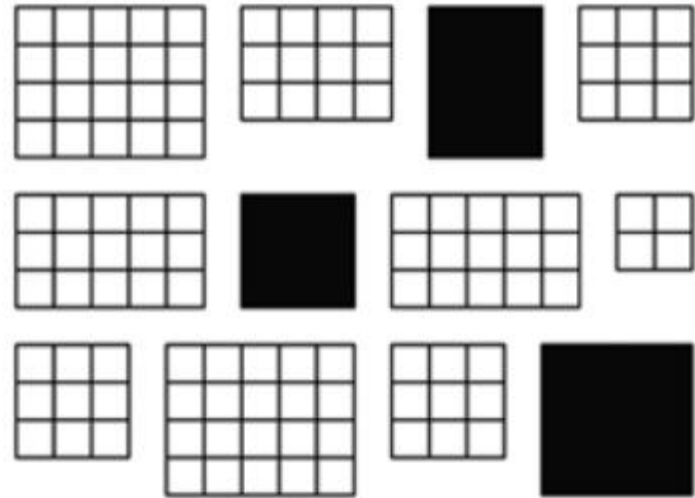
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Cluster Sampling

Take an SRS of clusters; observe all elements within the clusters in the sample.

-



Cluster sampling vs stratified sampling

Stratified Sampling	Cluster Sampling
The variance of the estimator of $\bar{y}_U$ depends on the variability of values <i>within</i> strata.	The cluster is the sampling unit; the more clusters sampled, the smaller the variance. The variance of the estimator of $\bar{y}_U$ depends primarily on the variability <i>between</i> cluster means.
For greatest precision, individual elements within each stratum should have similar values, but stratum means should differ from each other as much as possible.	For greatest precision, individual elements within each cluster should be heterogeneous, and cluster means should be similar to one another.



Cluster sampling vs stratified sampling

- Stratification generally increases precision when compared with simple random sampling, cluster sampling generally decreases it.
- Members of the same cluster tend to be more similar than elements selected at random
- By sampling everyone in the cluster, we partially repeat the same information instead of obtaining new information, and that gives us less precision for estimates of population quantities.
 - E.g., 100 individuals from a cluster may not provide the same amount of information as 100 independent individuals

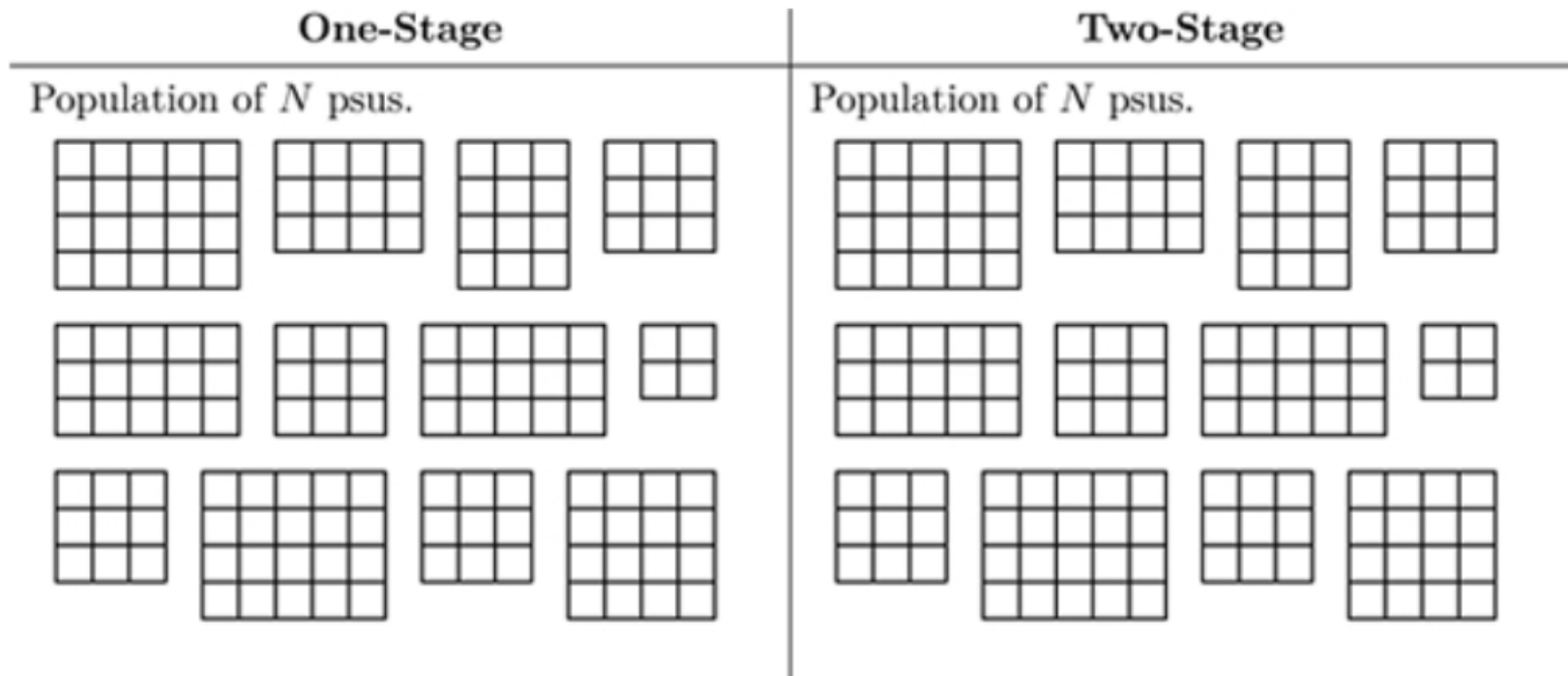


When clustering is ignored

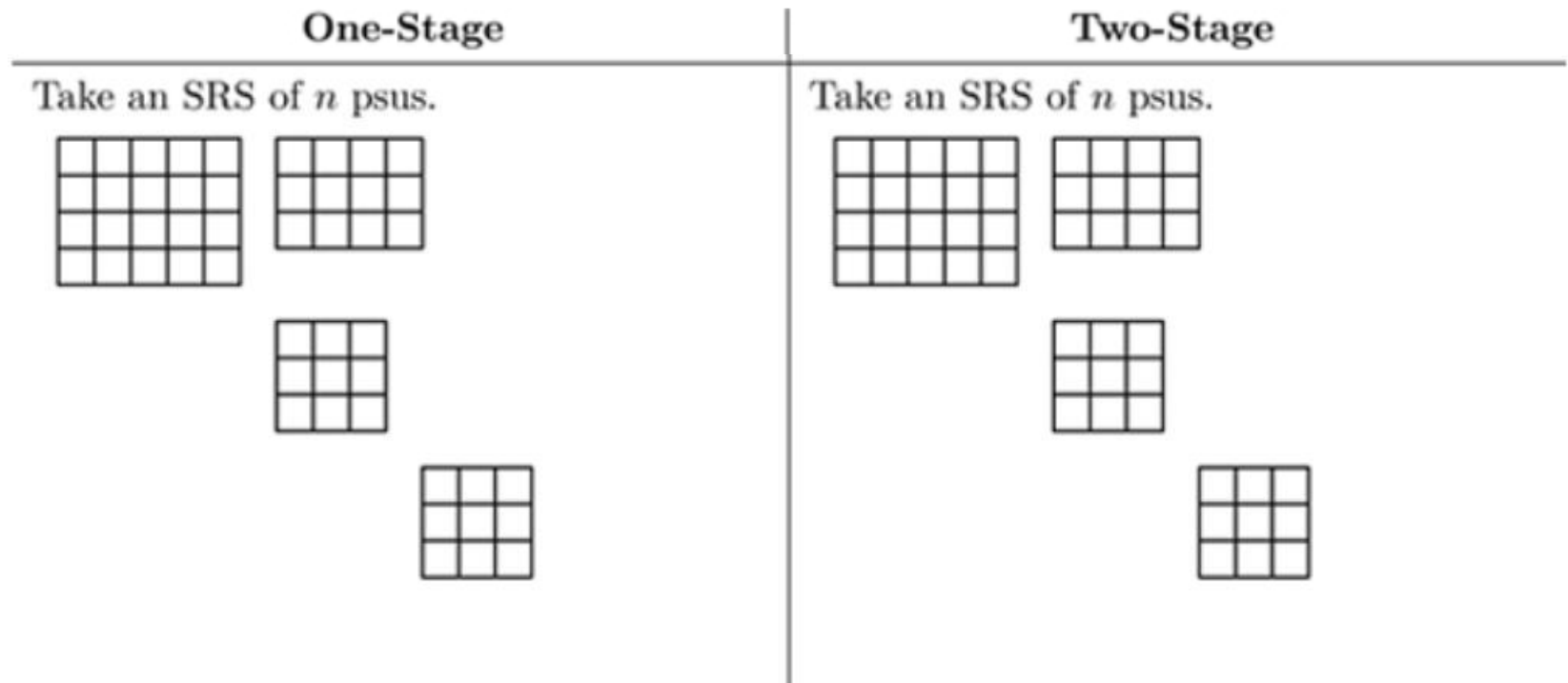
- Individuals in a cluster are more similar so their responses are not independent
- If ignoring clusters and using traditional statistical methods, standard errors would be underestimated
- Results appear more accurate than they are



One-stage and two-stage cluster sampling



One-stage and two-stage cluster sampling



One-stage and two-stage cluster sampling

